GuIA: Renaixement Museum Interactive Audioguide

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Abstract GuIA is a web-based interactive audioguide developed for the Museu del Renaixement in Molins de Rei. The system combines a conversational large language model with a RAG system and a curated Neo4j Knowledge Graph to generate museum explanations based on institutional data. Unlike a conventional audioguide with fixed recordings, GuIA allows visitors to ask questions and adapts its responses based on the selected language, explanation style, and accessibility preferences. This paper explains the architecture and implementation of the deployed prototype. To assess system performance, we introduce a quantitative evaluation framework covering four components: (1) retrieval quality, measuring the system's ability to access relevant knowledge graph information; (2) generation faithfulness, evaluating whether responses are grounded in retrieved data and free from hallucinations; (3) cultural bias deviation, assessing potential distortions in different languages in the presentation of historical content; and (4) explanation style evaluation, analyzing the system's adaptability to different user profiles. However, given the interactive and user-centered nature of this technology, and the aim to make it accessible, user evaluation remains central to validating its effectiveness.

Keywords Artificial Intelligence, Large Language Models, Retrieval-Augmented Generation, Knowledge Graphs, Interactive Audioguide, Museums, Accessibility, Cultural Heritage.

Github Repository https://github.com/L3ms123/GuIA
HuggingFace Deployment https://huggingface.co/spaces/RafelSV/guia-demo

1. Introduction

Museum audioguides are commonly designed as a fixed collection of recordings or written descriptions. This approach is reliable because the content is prepared in advance, but it offers limited flexibility. Visitors may have different interests, levels of prior knowledge, language preferences, and accessibility needs. They may also want to ask questions that were not anticipated when the guide content was written.

GuIA is an interactive conversational audioguide developed for the Museu del Renaixement in Molins de Rei. Conceived as a digital mediation device, it connects visitors with the collection by making objects, spaces, and historical contexts more understandable and meaningful, adapting its explanations to each visitor's language, profile, and accessibility needs. A visitor can select a preferred language and explanation style, indicate a room or artwork, type or dictate a question, and receive a grounded response in written and spoken form. GuIA is currently a browser-based web application with no offline application mode.

The system is built on a Retrieval-Augmented Generation (RAG) architecture that grounds every answer in a curated museum Knowledge Graph, ensuring factual reliability without sacrificing conversational flexibility. When information is not available, GuIA refuses to speculate rather than risk introducing errors. Accessibility is treated as a first-class design principle: configurable interaction modes, multilingual support, and a dedicated Easy-Read pipeline are integrated from the outset rather than added after the fact.

The main contributions of this paper are:

1. a web-based museum audioguide supporting conversational text and voice interaction, accessible directly from the visitor's smartphone browser;
2. a response pipeline that grounds visitor questions in a curated, updatable Knowledge Graph using retrieval-augmented generation;
3. a contextual adaptation mechanism that dynamically tailors responses based on interaction context and user needs;
4. an accessibility-first Easy-Read generation pipeline integrated via a separately deployed iDEM-based API;
5. a framework for assessing retrieval recall, generation faithfulness, cultural bias, and prompt sensitivity in a real museum context.

2. Design Principles of GuIA

Four core principles guide both the technical architecture and the interaction design of GuIA.

Curatorial control over generation. All responses must remain grounded in verified museum knowledge. The system does not allow free-form generation from the language model's own training data. Content authority belongs to the museum, and the Knowledge Graph is the mechanism through which that authority is enforced at inference time.

Refusal over hallucination. When the Knowledge Graph does not contain sufficient information to support an answer, GuIA explicitly declines to respond rather than producing a plausible but unsupported claim. Transparency about the limits of the system is treated as more valuable than apparent completeness.

Accessibility as a first-class concern. Interaction modes, language options, and explanation styles are designed from the outset to accommodate diverse visitor needs, including visitors with visual, cognitive, linguistic, or sensory barriers. Accessibility is not a compliance checkbox but a design constraint that shaped the architecture of the interface, the response pipeline, and the onboarding process.

Evaluation beyond technical accuracy. The system is assessed not only on retrieval and generation correctness, but also on usability, accessibility, and the quality of visitor understanding. A correct answer that a visitor cannot understand or access does not fulfil the system's purpose.

3. Project Definition

GuIA responds to a concrete communication challenge at the Museu del Renaixement. The museum receives visitors with different ages, languages, levels of prior knowledge, and accessibility needs. A single fixed explanation cannot serve all these profiles equally well. This gap is particularly acute because the current visit experience depends strongly on Catalan-language mediation, while a meaningful share of the public comes from outside Catalonia or is not comfortable following historical explanations in Catalan Museu del Renaixement (2024).

From the museum's perspective, the problem is not simply to digitise an audioguide. The museum needs a tool that preserves curatorial control while making the collection more accessible to visitors who arrive with different questions and different forms of access. GuIA therefore treats the audioguide as a mediation device: a system that connects the visitor, the museum space, and the museum's validated knowledge through an adaptive interface.

The objective of the project is to build and evaluate a browser-based prototype that provides grounded, multilingual, and accessible explanations during the visit. The system should allow a visitor to identify a room or artwork, ask questions in natural language, receive answers based on museum-validated information, and adapt the form of the explanation to their preferences. The project also aims to give the museum a maintainable structure for updating content and monitoring system behaviour over time.

A. Ethical, Legal, and Social Considerations.

GuIA operates in a context where accessibility, data protection, and institutional responsibility are central concerns.

Accessibility obligations. As a public-facing museum service, the system is designed and reviewed against WCAG 2.2 Level AA: content must be perceivable, operable, understandable, and robust. This is especially important given that GuIA explicitly addresses visitors who may face visual, cognitive, linguistic, or sensory barriers.

Data protection. GuIA applies a data-minimisation approach and does not ask visitors for names, accounts, contact details, or other direct identifiers. Preferences, a randomly generated session identifier, and recent chat messages are stored locally in the visitor's browser to preserve the session after a reload. Server-side analytics are disabled by default; when enabled for controlled testing, they store interaction metadata and random visit and session identifiers, but not the text of visitor questions or generated answers. These identifiers can link events from the same browser, but they are not intended to identify a named visitor or create an individual visitor profile. During onboarding, GuIA explains that text and audio are processed to provide the service and advises visitors not to disclose sensitive personal information.

AI transparency and oversight. GuIA informs visitors during onboarding that they are interacting with an AI-based system and that generated answers may contain errors. Although GuIA does not fall into a high-risk category under the EU AI Act, the design maintains transparency and human oversight. The museum retains editorial authority: it validates the Knowledge Graph, reviews sensitive outputs, and corrects errors. The system is not treated as an autonomous authority over the interpretation of the collection.

Social impact and inclusion. The primary social benefit of GuIA is its potential to reduce barriers for visitors who are not well served by a conventional audioguide, including non-Catalan speakers, visitors with disabilities, older visitors, and those with limited prior cultural knowledge. Realising this benefit requires careful deployment: GuIA should complement human guides rather than replace them, and it must ensure that personalisation does not become fragmentation. Visitors may receive explanations in different styles, but the factual basis and cultural sensitivity of those explanations must remain consistent.

Cultural representation. Language models may privilege dominant narratives. The Knowledge Graph is therefore not only a technical database but also a cultural document: its content, categories, and omissions shape how the museum is explained to visitors. Cultural bias audits and curatorial review are part of the long-term responsibility that comes with deploying the system.

4. Related Work

A. Conventional Museum Audioguides.

The evolution of institutional museum interpretation has historically relied on automated electronic devices to manage growing visitor volumes and overcome the scarcity of traditional human docents Ruiz et al. (2011). Conventional museum audioguides represent a mature paradigm in distributed cultural education, with millions of devices deployed globally on a daily basis Ruiz et al. (2011). These systems are highly valued by curators because they deliver carefully authored, authoritative descriptions that preserve strict historical context and objectivity Ruiz et al. (2011). Early technological advancements by organizations like GVAM pioneered portable multimedia configurations such as Ultra-Mobile Personal Computers (UMPCs) to deliver multi-language tracks and enrich the self-guided exploration of collections Ruiz et al. (2011) Mora (2025).

Despite their prevalence, conventional audioguides suffer from systemic architectural rigidity. They impose predefined, linear itineraries that restrict spontaneous user autonomy and fail to adapt to real-time visitor interests or immediate moods Ruiz et al. (2011). Because these legacy systems rely on hardcoded, fixed-content tracks, they function exclusively as one-way broadcasting systems and are entirely incapable of answering open, spontaneous questions posed by the visitor Ruiz et al. (2011).

B. Conversational Agents for Cultural Heritage. To satisfy modern cultural consumption habits, museums have shifted from presenting passive exhibitions to fostering environments of active participation and discovery Ruiz et al. (2011). This evolution has driven the development of conversational agents specifically optimized for cultural heritage spaces. Unlike static interfaces, conversational agents provide personalized interaction membranes that allow users to "ask rather than listen" and interrogate exhibits at their own pace Ruiz et al. (2011).

Modern conversational frameworks leverage NLP to break down information silos across fragmented museum databases, integrating structured catalog records, imagery, and raw descriptions into unified systems Li et al. (2026). Platforms such as MuseKG demonstrate how interactive knowledge graph architectures can map complex relationships between historical figures, locations, and artifacts, exposing a lightweight natural language interface that translates open-ended user queries into precise data lookups Li et al. (2026). By engaging visitors in dynamic, multi-turn dialogues, these agents allow the exploration of interconnected historical narratives that go far beyond the scope of traditional static text panels.

C. Retrieval-Augmented Generation (RAG) and GraphRAG Systems. While general-purpose LLMs offer unparalleled flexibility in text generation, their direct application to highly specialized cultural domains introduces critical risks. LLMs operating in isolation are prone to factual hallucinations, suffer from fixed knowledge cutoffs, and behave as uninterpretable "black boxes" that lack domain specificity Wang and Zhang (2025). To mitigate these errors, Retrieval-Augmented Generation (RAG) architectures ground LLM generation by supplying an external, curated database of evidence from which the model must pull its answers, thereby constraining the output to verified institutional knowledge.

Traditional text-based RAG relies primarily on vector databases to execute semantic similarity searches across disjointed document chunks, a mechanism that frequently struggles to capture the complex, highly interconnected relational narratives inherent in certain contexts like cultural heritage data. To bypass this limitation, Knowledge Graph-based RAG (KG-RAG) and GraphRAG systems tightly couple LLMs with structured knowledge graphs Wang and Zhang (2025). As established by Microsoft Research, the GraphRAG framework uses language models to extract global semantic networks from unstructured narrative data, creating hierarchical community summaries that maximize information retention and provenance tracking Larson and Truitt (2024). Furthermore, anchoring generative frameworks to formalized ontologies (such as CIDOC CRM) enforces strict structural consistency during inference, guaranteeing that the retrieved context maintains clear, logical relationships aligned directly with the institution's base taxonomy Wang and Zhang (2025).

D. Accessible Museum Technologies, Easy-Read Research, and Visual Guidelines. A core requirement of modern museography is the "Design for All" principle, which mandates that multimedia devices provide universal accessibility so they can be seamlessly utilized by individuals regardless of their sensory or cognitive capabilities Ruiz et al. (2011). In early multimedia guide frameworks, such as GVAM's foundational accessible prototypes developed alongside organizations like CESYA, ONCE, and CNSE, this was accomplished by standardizing specific interface layouts Ruiz et al. (2011). For users lacking access to the audio tracks, systems integrate permanent, synchronized subtitle boxes and floating, draggable sign-language windows Ruiz et al. (2011). For visitors with visual impairments, systems deploy audio-guided navigation, screen magnification, and contrast modifiers Ruiz et al. (2011).

Modern accessible museum technologies have expanded these boundaries by incorporating:

- **Easy-Read Research:** This methodology focuses on simplifying the linguistic structure, vocabulary, and syntactic complexity of curatorial content, ensuring that deep historical concepts are cognitively accessible to individuals with learning difficulties, neurodivergent conditions, or language barriers.
- **Visual-Description Guidelines:** These protocols provide a structured, standardized vocabulary to systematically translate visual artwork into verbal or textual descriptions, ensuring that spatial layouts, colors, textures, and artistic techniques are accurately and expressively conveyed to blind or low-vision visitors.

E. The Technical Distinction of GuIA. The architectural limitations of existing museum systems highlight a sharp dichotomy in cultural heritage technologies. Traditional fixed-content audioguides provide carefully authored descriptions that accurately safeguard a museum's institutional knowledge and curatorial framing, but they cannot answer open questions Ruiz et al. (2011). Conversely, general-purpose conversational assistants can fluidly answer open questions, but they operate without structural boundaries, frequently hallucinating information and failing to respect the strict historical and educational boundaries mandated by curators Wang and Zhang (2025).

GuIA directly resolves this tension by combining a situated museum context with open conversational generation GVAM Guías Interactivas (2025). Rather than forcing the user into a rigid audio script or leaving them to interact with an ungrounded general-purpose chatbot, GuIA achieves conversational flexibility by anchoring its generation engine directly to a retrieval loop over a curated, museum-specific graph structure. This synthesis ensures that while visitors enjoy the cognitive freedom of open dialogue, every response remains rigorously constrained within the authentic, verified institutional parameters of the museum.

5. System Overview

GuIA is divided into a web interface, a backend service, a Neo4j Knowledge Graph, an audio module, and a separately hosted iDEM-based service for Easy-Read generation. The application has been deployed on Hugging Face Spaces so that it can be accessed through a browser without installing an application. Figure 1 summarizes the main components of the system and their data flow.

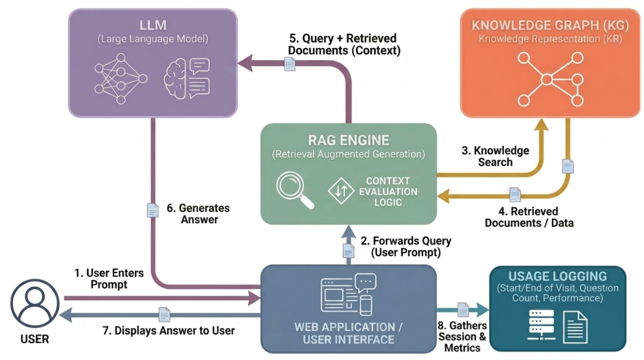

Fig. 1. Overview of the GuIA system architecture.

A. Overall Architecture. The general workflow is:

1. The visitor opens the web interface and completes the onboarding process.
2. The visitor selects a language, guide style, optional age range, and accessibility preferences.
3. The visitor selects a room or artwork manually, scans a QR code, or asks a question without selecting a location.
4. The backend uses the question and the current museum context to retrieve relevant information from the Knowledge Graph.
5. The retrieved museum information is sent to the language model together with the visitor's preferences.
6. The system generates a personalized answer and returns it to the visitor as text and, when selected, as spoken audio.

This architecture separates museum knowledge from natural-language generation. Neo4j stores the curated content, while the language model is used to identify relevant information and transform it into an understandable explanation. This helps preserve the museum's control over the factual basis of the answers.

B. End-to-End Request Flow. A typical interaction begins in the frontend. The visitor completes onboarding, which creates or reuses a browser session identifier and stores the selected language, guide style, age profile, accessibility preferences, and current location in local storage. When the visitor selects a room or artwork manually, through a QR code, or through a NaviLens marker, the frontend sends the selected context to the backend so that it can be attached to later questions.

When the visitor asks a question, the chat module sends the message, session identifier, language, profile, location, and accessibility flags to the streaming chat endpoint. The backend first combines the explicit request with the stored session context. It then retrieves relevant rows from Neo4j, using the current room or artwork to disambiguate situated questions such as "What is this?" or "Who made it?". The retrieved rows and generated Cypher query are inserted into the system prompt together with the visitor preferences. The answer is then generated and streamed back to the interface. If spoken audio is enabled, complete sentences are sent to the audio service so that playback can begin before the full response has finished.

When Easy-Read mode is active and Neo4j returns usable context, the backend routes the response through the iDEM-based service. If iDEM is unavailable or fails, the system falls back to the normal Cohere generation path with simple-language instructions. This keeps the interaction available while making the fallback visible in the backend event source.

C. Backend API Surface. The frontend communicates with a small set of backend endpoints. The main conversational route is `/chat/stream`, which returns server-sent events for streamed answers. A non-streaming `/chat` route is also available. The `/context` route stores the current room and artwork for a session, while `/locations` returns the room and artwork catalogue used by the location selector. The `/translate-conversation` route translates visible conversation items when the visitor changes language, and `/easy-words` supports simplified-word assistance.

Audio is handled by a separate service during local development. The `/transcribe` route receives recorded microphone audio and uses Whisper to return text. The `/transcribe/warmup` route preloads the Whisper model to reduce first-use latency. The `/speak` route converts generated text into an MP3 response using `edge-tts`. In the Hugging Face deployment, these services are served from the same public application address; during local development, the frontend points to separate local ports for the LLM and audio APIs.

The application also exposes metadata-only analytics through `/analytics`. Administration routes are protected separately under `/admin/api/`. They are not part of the visitor workflow and require the configured admin password.

6. Frontend Architecture

A. Web-Based Interface. The interface is implemented with HTML, CSS, and JavaScript. No native mobile application installation is required. The web application is intended to be opened from a smartphone browser during a museum visit.

The frontend includes the onboarding process, the chat interface, language selection, accessibility options, audio playback, voice recording, QR scanning, location selection, and an initial tutorial. These elements have been refined through several design iterations.

B. Onboarding and Visitor Preferences. At the start of a session, the visitor completes a three-step onboarding process. The first step configures:

1. The interface and response language: Catalan, Spanish, or English;
2. The guide explination styles:
    (a) Normal guide: prioritizes the most important information.
    (b) How the artwork was created: Explains colors, materials, and how the artist worked.
    (c) Story-style explanation: Explains the visit as a simple story.

(d) Learn more details: Provides additional historical information and context.

3. An optional age range: child/teenager, young adult, adult, or senior.

The second onboarding step contains accessibility options:

1. Larger text;
2. Uppercase text;
3. Visual descriptions of artworks;
4. Simpler language;
5. Spoken responses;
6. Slower speech;
7. An optional initial tutorial.

The third step displays an AI and privacy notice. GuIA explains that the guide uses artificial intelligence, that generated information can contain errors, and that text or audio is processed when the corresponding interaction mode is used. Before entering the main interface, the visitor confirms that this notice has been read.

C. Session Persistence. The browser stores the selected preferences and the current session. This allows the interface to preserve the visitor's settings and museum location if the page is reloaded. The backend also keeps the recent conversation context so that the guide can understand follow-up questions.

For example, a visitor may first ask about an artwork and then ask, "Who created it?" or "Can you tell me more?" The second question does not repeat the artwork title, but GuIA can use the previous interaction to understand what the visitor is referring to.

D. Location Selection and NaviLens Codes. GuIA supports both manual and code-based location selection. A visitor can manually select a museum room and, optionally, a specific artwork. The selected location is sent to the backend and used as contextual information when generating explanations.

The system also uses NaviLens codes to identify rooms and artworks. NaviLens is an accessible signage technology designed to be detected from a greater distance and without requiring precise camera alignment. This is particularly useful for blind and low-vision visitors, who can scan the codes using the NaviLens application and receive accessible navigation support.

Each NaviLens code can also include a conventional QR code. This allows the same physical marker to be used by different visitors: sighted users can scan the standard QR code with GuIA's built-in camera scanner, while blind and low-vision users can use the NaviLens application. Both interaction paths provide GuIA with the corresponding museum location, allowing the system to adapt its explanations to the room or artwork being visited.

E. Streaming Conversation. The answer is sent progressively to the interface while it is being generated. This reduces the perceived waiting time and makes the interaction feel more natural. The interface displays a writing indicator until the first part of the answer arrives. When spoken audio is enabled, the guide can begin reading complete sentences while the rest of the response is still being prepared.

F. Iterative Development of the Interface. The design and development of GuIA followed an iterative process grounded in user feedback, expert input, and co-design with accessibility organisations. The objective was not only to make the guide easier to understand and use inside the museum, but to ensure that accessibility options were integrated into the core experience rather than added as a separate layer — a principle directly stated by Estel · la Oncins Noguer in the expert interview: *"Accessibility should be considered as a transversal aspect [...] a museum, when planning the launch of a solution, should already consider questions related to accessibility from the beginning."*

First iteration: personalisation and onboarding. The first version focused on the basic onboarding process. Visitors could select a language, an age group, and the type of explanation they preferred. This established the core design principle of GuIA: different visitors should not necessarily receive the same explanation. The interface was subsequently reorganised into a clearer three-step flow. Step indicators were added so that visitors could understand their progress, and the accessibility settings and privacy notice were separated from the initial personalisation options to reduce cognitive load and avoid overwhelming users at the start of the session.

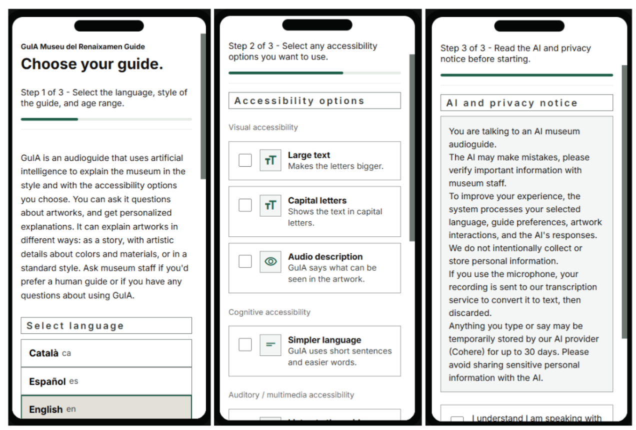

Fig. 2. Onboarding interface of GuIA, showing the three-step configuration process for personalisation, accessibility options, and AI/privacy notice before entering the main interface.

Second iteration: voice interaction and audio controls. A second iteration introduced voice interaction. Visitors could ask questions using the microphone and listen to the answers instead of relying only on written text. This change was motivated by two converging concerns identified in the expert interviews. The accessibility specialist highlighted that traditional audio guides assume visual access to the artwork, meaning that a visitor who is not looking at the object also misses the information typically anchored to it. The museum director emphasised that the interface should complement rather than compete with the artworks: *"Technology and artificial intelligence cannot and must not hide the importance of the original artwork. It has to work in parallel."* Supporting voice output allowed visitors to listen to explanations while maintaining their attention on the collection, rather than reading from a screen.

Audio controls were progressively improved with options for slower speech, volume adjustment, and the ability to pause and resume playback. Larger text was added as a visual accessibility option. These changes were particularly relevant because GuIA is designed to be used while moving through a museum, where reading from a screen may not always be practical.

Third iteration: location selection and NaviLens integration. The location-selection process was revised several times. Initially, visitors selected their location manually from a floor and room selector. QR scanning was later added so that a room or artwork could be identified directly with the smartphone camera. The system was subsequently connected with NaviLens codes, a high-density optical code standard designed to be detectable from a greater distance and without precise camera alignment. This integration was specifically motivated by the needs of blind and low-vision visitors, as recommended by the Asociación Discapacidad Visual Catalunya B1+B2+B3, whose specialists emphasised the importance of supporting multiple access modalities and full compatibility with assistive technologies.

Each NaviLens marker also encodes a conventional QR code, so the same physical signage serves sighted visitors using GuIA's built-in scanner and blind or low-vision visitors using the NaviLens application directly. The session is preserved when the location changes, allowing the visitor to continue the conversation without restarting the guide.

Fourth iteration: accessibility options and co-design revisions. As accessibility requirements became clearer through the co-design process, the set of configurable options was expanded. Accessibility settings were reorganised into three categories — visual, cognitive, and auditory — to make the onboarding choices easier to navigate. These categories now include large text, uppercase text, audio description, simplified language, spoken audio, and slower voice speed.

A concrete revision emerging from the co-design process with Aura Fundació was the replacement of decorative images used as interface elements with pictograms. Specialists from Aura Fundació identified pictograms as more easily and unambiguously interpretable for users with intellectual disabilities, and this change was implemented before the preliminary user testing session with Down Tarragona. Participants in that session responded positively to the pictograms and reported that the interface was understandable, though some interaction steps still required effort. The session also revealed that the system was not sufficiently robust to grammatical errors and incomplete queries, and that some Easy-Read outputs remained too complex — findings that informed subsequent improvements to query normalisation and the simplification pipeline.

Tutorial modes. As the interface grew more complex, a structured onboarding tutorial became necessary. An initial help dialog was introduced to explain the main controls. This was later replaced by two dedicated tutorial modes, introduced following the recommendation of the accessibility specialist to support users who might not be familiar with conversational AI guides in a museum context and to reduce confusion during first use.

The visual tutorial is an interactive step-by-step guide displayed as a modal dialog. It highlights interface elements sequentially and waits for the visitor to perform the required action before advancing, so that the flow matches the visitor's own pace. The spoken tutorial is a screen-reader-compatible alternative that presents simplified instructions in an accessible overlay and can optionally read them aloud, supporting visitors with visual impairments who may not use a dedicated screen reader. Both modes can be activated during onboarding or accessed later from the help button in the main interface, and neither assumes that the visitor already knows what a conversational audioguide is.

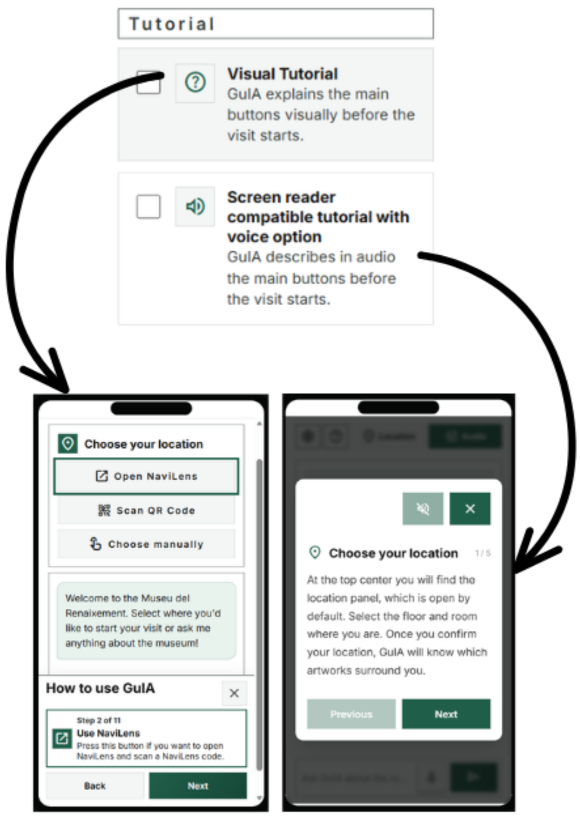

Fig. 3. Tutorial screens showing the visual and spoken guidance modes available in GuIA.

Fifth iteration: mobile usability and screen-reader compatibility. A later group of iterations focused on mobile usability and screen-reader compatibility. The interface was adjusted for tablets and smaller mobile screens. Screen-reader announcements were added for all interactive actions, including opening the location panel, scanning a code, and changing settings. The co-design collaboration with the Asociación Discapacidad Visual Catalunya B1+B2+B3 directly informed these changes: their specialists recommended full screen-reader compatibility for users who rely on assistive technologies, as well as integrated audio output for individuals with low vision who do not use screen readers but still benefit from auditory support. A WCAG 2.2 compliance review was conducted in parallel, using a combination of automated tools and manual checks to verify that contrast, keyboard navigation, ARIA labels, and live regions met the applicable requirements.

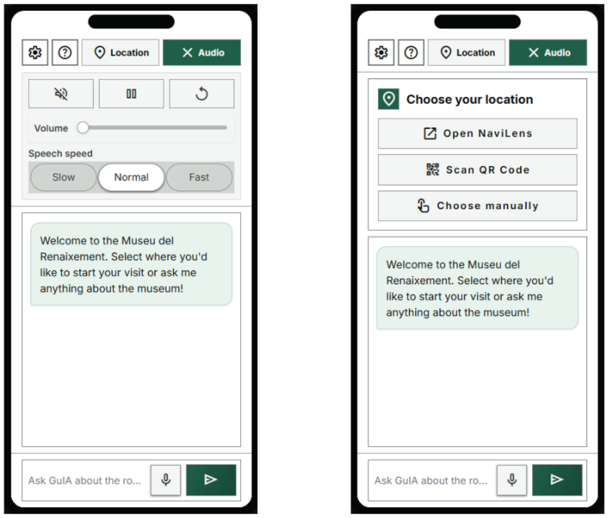

Fig. 4. GuIA's main mobile chat interface, showing the chat panel, audio controls, and location panel.

Frontend architecture. As the number of functions increased, the frontend code was divided into separate modules for onboarding, audio, voice recording, chat interaction, location selection, translations, session persistence, and accessibility behaviour. This modular structure made it possible to revise individual components in response to feedback — for example, adjusting the transcription display to show the recognised text in the input field before sending, a decision motivated by the finding from the Down Tarragona session that speech recognition may produce errors with atypical speech patterns and should not be treated as perfectly reliable.

Administrative tools and long-term maintainability. Finally, the project added tools to support long-term use by the museum. Optional analytics were implemented to record interaction patterns — including navigation paths, frequently accessed artworks, and selected language preferences — without storing the text of visitors' conversations, in line with the data minimisation principles discussed in the regulatory section. A password-protected administration interface was added so that museum staff can update the Knowledge Graph content, review unanswered visitor queries, and download anonymised usage data without requiring direct database access.

7. Knowledge Graph Layer

A. Data Used for Knowledge Graph. For this project the Museum provided us with several types of data:

- A .xlsx file that contains information about each artwork that has ever been displayed in the Museum separated into three different tabs, past artworks, current artworks, and future artworks. Each tab contains artwork name, author, location in the museum, technique used to create the artwork, and physical location within the museum.

- A docx. document with an audio guide written by the museum. Contains Information about the museum, each room theme and artworks, as well as their history.

- A docx. document with a short desription of each room and section in the museum.

After inspecting the data, in the course of the development of the app, we have decided to add additional data from different sources not provided by the museum to fill the database:

- Museum website information with additional artwork descriptions and their precise location, including photos.

- Wikipedia information about different authors, technique used and historical data

- Generated text of visual descriptions from the pictures in the website. It was made to make text of visual descriptions of each artwork for people with vision problems, to describe information like texture, details, the foreground of the artwork etc.

This additional information helped us to make the Knowledge Graph complete and can always be replaced with confirmed information at any moment without issues.

B. Knowledge Graph Development. From the start of the development we have decided to make a relational database for an easy retrieval of the data without augmentation by LLM.

First try of the development was to try and create the Knowledge Graph using only LLM Graph Constructor.

For this we have used Neo4j LLM Knowledge Graph Builder, without the predefined schema, we have fed the Graph Builder the information we had and the result was disappointing. The Graph builder made hundreds of thousands of nodes with repeating relationships using different wordings (Examples "Made_by" "Created_by" "Invented_by" "Sculptured_by").

In our second attempt we tried to add the predefined schema to the Graph builder, but the results were disappointing as well. Even with schema the Graph Builder have created a lot of repeating and unnecessary nodes with lackluster connections.

Our third attempt was a final solution. We have discarded the idea of using LLM Graph builder to generate new nodes and instead we made a parser which filled our starting schema in neo4j database, which contained 3 nodes: ArtPiece, Artist, Technique. This gave us a direct control of how we create our nodes and made the process of creating of new nodes from the .xlsx fast. This Graph also contained only 2 connections, being CREATED_BY, USED_TECHNIQUE for connecting ArtPiece's with Artist and Technique nodes.

We have decided to use neo4j for it's simplicity and ease of integration. Having relationships between nodes be represented in natural language is easy to navigate for RAG which uses natural language from the user as the primary input. For example if the users says "Who was the author of this artwork". the RAG can translate this easily into an actual query `MATCH (a:ArtPiece)-[:CREATED_BY]->(ar:Artist)`. For the main data storage we have decided to host the data on the neo4j cloud server neo4j aura.

Our next step was to expand the Graph with additional information. First was to make a location nodes, to identify the location of nodes by rooms and floors and make the location based search faster. We have added Sala and Planta nodes, which were connected to all the artworks in the museum with the connection LOCATED_IN.

For visual descriptions we have created a separate node type VisualDescription, containing information about artworks looks and feelings. Additional the node Theme was created, it contained short thematic description which may apply to several artworks, it was created to help us with search based on the users artwork thematic interpretation.

Our last step was to make the node to store metadata about questions. The app calls the neo4j database each time the answer is provided or not and stores the data in the nodes with names AdminUnresolvedQuestion/AdminAnalyticsEvent. When the user makes a question we collect data such as average question length, language used, average question asked an so on. We collect it in AdminAnalyticsEvent node. When there is no answer we store the unresolved question in AdminUnresolvedQuestion. These nodes have no relationships and serves only as storage and cannot be retrieved by a regular user (Used for museum analytics).

C. Knowledge Graph Structure and Statistics.

Node	Main stored information
`ArtPiece`	`artwork_id`, `title`, `artist`, `dating`, `technique`, `description`.
`Artist`	`name`, `biography`.
`Technique`	`name`, `description`.
`VisualDescription`	`artwork_id`, `visual_overview`, `audio_description`, `colors`, `materials`.
`Sala`	`id`, `palau`, `info`, `path_to_sala`.
`Planta`	`id`, floor metadata.
`Theme`	`name`, `description`.
`AdminAnalytics`	`id`, `event`, `recordJson`, `ts`.

Relationship	Use in the system
`ArtPiece` → `Artist`	`CREATED_BY`: Links artwork to its creator.
`ArtPiece` → `Technique`	`USES_TECHNIQUE`: Links artwork to its technique.
`ArtPiece` → `Sala`	`LOCATED_IN`: Connects artwork to display room.
`ArtPiece` → `VisualDescription`	`HAS_VISUAL_DESCRIPTION`: Accessibility data retrieval.
`Planta` → `Sala`	`HAS_SALA`: Represents spatial organization.
`Planta` → `Theme`	`FEATURES_THEME`: Connects areas with themes.
`Artist` → `Artist`	`CONTEMPORARY_OF`: Links artists from same period.

Here you can see all Nodes, Relationships and Key Properties of nodes present in the Knowledge Graph.

Total nodes amount: 1240 Total relationships count: 250

From	Relationship	To	Count
ArtPiece	LOCATED_IN	Sala	70
ArtPiece	USES_TECHNIQUE	Technique	70
ArtPiece	HAS_VISUAL_DESCRIPTION	VisualDescription	70
ArtPiece	CREATED_BY	Artist	27
Artist	CONTEMPORARY_OF	Artist	6
Planta	HAS_SALA	Sala	5
Planta	FEATURES_THEME	Theme	2

This graph shows us the amount of each type of relationships. As we can see from it:

- Each ArtPiece has a location, technique used, artist and visual description (Total of 70 artworks). So the information about each artwork is complete.
- Each floor has a room in it. It helps us with navigating the graph using location based queries from the user (Example: "I'm in Room 2 and I see this artwork, can you tell me more about it")
- Each Floor has a theme

Next let's look at our unique nodes count and their distribution:

Label	Node Count
AdminAnalyticsEvent	982
ArtPiece	70
VisualDescription	70
Technique	40
Theme	38
Artist	24
AdminUnresolvedQuestion	7
Sala	5
Planta	3
MuseumInfo	1

- The majority of nodes are AdminAnalyticsEvent, used for storing json information about asked questions. This type of nodes is ever growing, since it is automatically updated every time some question is asked.
- Then we have nodes in the double digits: ArtPiece, VisualDescription, Technique, Theme, Artist. They are all connected and serve as the primary information storage for all the Artistic Data.
- Lastly we have AdminUnresolvedQuestion, Planta, MuseumInfo, Sala. AdminUnresolvedQuestion is used to store unresolved questions, other nodes are permanent information that will likely not change.

Out of all of these nodes AdminAnalyticsEvent and AdminUnresolvedQuestion cannot be accessed by RAG, it is strictly for the museum usage and analytics.

At the end let's look at the complete visualization of our graph:

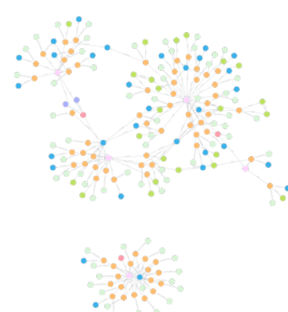

Fig. 5. Full Knowledge Graph.

The graph is to big to show in detail, the lines are different connections and the colors represent:

- Orange - ArtPiece
- Green - Artist
- Lime - VisualDescription
- Pink - Sala
- Red - Planta
- Blue - Technique

D. Interaction between the App Layers. This Graph interacts with several layers of our application.

- RAG Interaction - Next Section
- Direct creation of Analytics nodes from our Backend (neo4j API used to connect to the Graph and add new analytics nodes)
- Website Admin Page, used by museum workers to update all the relevant nodes and relationships using template csv files.

Admin page is a part of the final application that allows the museum workers to see the statistics from the AdminAnalyticsEvent nodes, update the nodes from Unresolved Questions page and update every node and relationship using csv and xlsx template files.

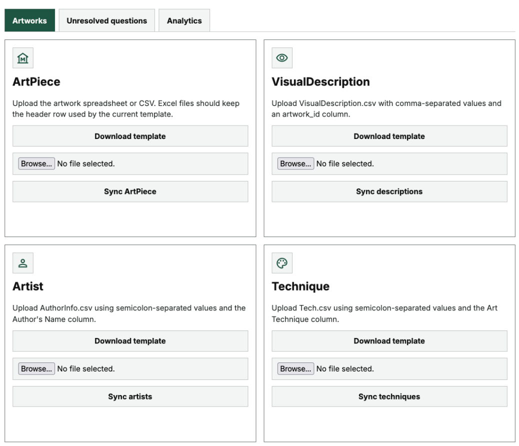

Fig. 6. Admin Graph Updater

On this image you can see the graph updater for the museum workers. It accepts the template of the right format and parses the data inside it, creating new nodes or update the existing ones. All the connections are created automatically, the script searches for the nodes of types Sala, Artist, Technique with similar properties and creates the relationship between them. Each type of node has its own template and automatic relationship creating script, since each node has a set of unique relationships.

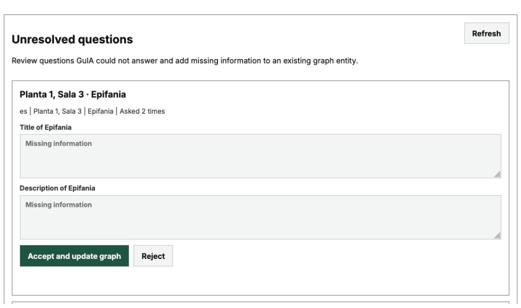

Fig. 7. Admin Unresolved Question Updater

Lastly the museum workers can directly see the unresolved questions and try to update the selected nodes. This was made in case some information can be quickly added, without having to make or update a whole csv/xlsx file.

8. Retrieval-Augmented Generation Pipeline

A. Retrieving Museum Information. When a visitor asks a question, GuIA first identifies the relevant museum information. The question is combined with the current room, the selected artwork, and the recent conversation context. This is important because many questions only make sense inside the museum space. For example, the question "What can I see here?" depends on the room where the visitor is standing.

GuIA uses Cohere's Command A model (`command-a-03-2025`) to translate the visitor's question into a Cypher query, which is the query language used by Neo4j. This query searches the Knowledge Graph for the relevant artworks, artists, techniques, locations, or historical information.

B. Generating the Explanation. Once the relevant information has been retrieved, it is included in the instructions sent to the language model. The model generates an explanation based on this museum context and the visitor's selected preferences. In this way, the system combines the flexibility of a conversational assistant with the reliability of a curated source of information.

The answers are written as short paragraphs so that they can be read comfortably in a chat interface and listened to as audio. The model is instructed to prioritize the most relevant information, use clear language, and avoid adding facts that are not supported by the Knowledge Graph.

C. Context and Location Handling. Location is treated as part of the question rather than as a separate navigation feature. The frontend keeps the current room and artwork in the browser session and sends updates to `/context`. The backend also stores this information in a server-side session context keyed by the session identifier. This allows follow-up questions to remain meaningful even when the visitor does not repeat the artwork name.

When a question is submitted, the backend uses the request payload first and falls back to the stored session context if the payload does not include a room or artwork. The retrieval step then receives the question, session identifier, current room, current artwork, and visual-description flag. This design is important because many museum questions are indexical: "this", "here", and "the painting" only make sense when the system knows where the visitor is standing. The previous graph context is also compacted and retained for follow-up retrieval, so a later question can refer back to the previous answer.

D. Failure and Fallback Handling. Several safeguards are implemented to keep the prototype usable when one component fails. First, generated Cypher is restricted to read-only graph access before execution. If the generated query fails, returns no useful rows, or is too brittle for the selected artwork title, the backend attempts a deterministic artwork-context fallback query based on significant title terms and, when available, the selected room. This fallback is designed to recover basic artwork context without allowing write operations.

Second, the generation prompt instructs the model to refuse unsupported claims. If Neo4j returns no rows, Easy-Read generation through iDEM is skipped because there is no grounded context to simplify. The system then uses the normal Cohere guide path with the same refusal and grounding rules.

Third, the Easy-Read service is optional at runtime. When simple-language mode is enabled and graph rows are available, GuIA first tries the iDEM guide route. If that call fails, the streaming endpoint falls back to Cohere with simple-language instructions and marks the source as `cohere-fallback`. For audio, the speech service returns an error if transcription or synthesis fails, and the frontend keeps the written interaction available. For transcription, Whisper also shows the transcribed text before submission, allowing the visitor to correct it when recognition is inaccurate.

9. Personalization and Accessibility

A. Language Adaptation. GuIA supports Catalan, Spanish, and English. The selected language is included as a strict rule in the generation prompt. The interface itself is translated using a frontend translation file. When the visitor changes language during a session, visible conversation items can be sent to the backend translation endpoint. The backend asks the LLM to translate each item while preserving order, role, meaning, names, and paragraph structure.

B. Guide Styles and Age Profiles. Guide styles affect the explanatory emphasis of the answer. The backend inserts a style-specific instruction into the system prompt. This allows the same retrieved museum facts to be phrased differently according to the visitor's preference.

The optional age profile is also included in the prompt. Its purpose is to adjust tone and level of explanation, not to infer personal characteristics automatically. The visitor explicitly selects the profile and may leave it undefined.

C. Visual-Description Mode. When visual-description mode is enabled, the retrieval query requests available visual, material, compositional, and spatial fields. The generation prompt instructs the model to produce a concise museum audio description:

1. identify the work using available label information;
2. summarize the subject, composition, and mood;
3. describe the work in a clear spatial order;
4. include concrete supported details such as figures, gestures, objects, colors, and textures;
5. introduce historical or social context after the visual overview.

The prompt explicitly forbids invented visual details. If the graph does not contain enough information for a fuller description, the answer should state this limitation. The effectiveness and appropriateness of generated descriptions still require human review and user testing.

D. Easy-Read Mode and iDEM. iDEM stands for *Innovative and Inclusive Democratic Spaces for Deliberation and Participation*. It is a European Horizon research project coordinated by Pompeu Fabra University. The project addresses language barriers that can exclude people from understanding complex information. Its work includes Natural Language Processing tools for detecting linguistic difficulty and adapting text to different comprehension needs.

We selected iDEM because its text-simplification approach is informed by linguistic expertise and is designed for accessibility-oriented language adaptation. This is particularly relevant for visitors who benefit from shorter sentences, common vocabulary, and clearer structure, including people with intellectual disabilities and other comprehension difficulties.

Starting from iDEM's repository, we developed our own API wrapper and deployed it as a separate Hugging Face Space. The separation is intentional: the main GuIA Space serves the museum application, while the iDEM Space loads and runs the simplification model. The GuIA backend calls the iDEM service through an HTTP endpoint.

When Easy-Read mode is activated, GuIA sends the visitor's question, the relevant museum information, the selected language, and the current room or artwork to this API. The API returns an explanation with shorter sentences, more familiar words, and a clearer structure while preserving the essential facts.

E. iDEM Service Implementation. The iDEM API is implemented with FastAPI and deployed independently from the main application. Its workflow is:

1. receive the visitor's question and the museum information retrieved by GuIA;
2. create an answer based on the available artwork information;
3. apply the iDEM simplification approach;
4. return the simplified explanation to the main application.

This design makes simplified language part of the conversational system. Visitors can still ask open questions and receive personalized answers. Formal Easy-Read content still requires human validation and evaluation with intended users.

10. Prompt Engineering: Technical Specification

A. Prompt structure. `build_system_prompt` assembles a dynamic system prompt from six components, following the structural taxonomy of Schulhoff et al. (2024):

1. **Role**: "You are GuIA, the AI audio guide of the Museu del Renaixement in Molins de Rei" — a zero-shot role prompt that anchors domain and persona.
2. **Directive**: task-level instructions covering scope (museum collection only), hallucination policy (refuse rather than invent), and conversation constraints (no follow-up questions, no field-absence apologies).
3. **Output formatting**: plain text, short paragraphs of 1–2 sentences, numbered lists for enumerations, no Markdown markers. These constraints are applied redundantly at generation time and enforced post-hoc via `sanitize_chat_text` and `format_chat_text` — an instance of the prompt engineering / answer engineering separation described in Schulhoff et al. (2024).

4. **Few-shot examples**: two fixed exemplars appended before the language rule. Example A demonstrates a grounded factual answer (artwork with title, artist, and dating retrieved). Example B demonstrates the correct refusal behaviour when graph rows are empty or irrelevant. Placement follows the survey's guidance that the most representative case should appear last, so Example B (the safety-critical no-information case) closes the block Schulhoff et al. (2024).

5. **Style instructions**: explanation style rules (*artist*, *storyteller*, *normal_guide*, *more_details*) and age-range descriptions are injected as `GUIDE STYLE` and `VISITOR PROFILE` fields, providing behavioural guidance on vocabulary depth, register, and assumed prior knowledge.

6. **Additional information (RAG context)**: retrieved Knowledge Graph rows and the generated Cypher query are appended as a structured `RETRIEVED NEO4J CONTEXT` block, followed by an `INTERNAL GROUNDING CHECK` directive instructing the model to verify each claim against the provided rows before generation and omit ungrounded content.

B. Conditional composition. Four prompt segments are appended conditionally at inference time based on request flags:

- `simple_language`: activates the Easy-Read segment specifying CEFR A1–A2 vocabulary targets.
- `visual_descriptions`: activates the audio-description segment specifying a five-step spatial sequence (label facts → overall impression → spatial order → material detail → historical context).
- `more_time`: activates the pacing segment requesting clear sentence boundaries for audio output.
- Detail detection: `user_requested_detail()` scans the message for explicit requests for more information and activates a richer 4–6 paragraph instruction when triggered.

C. Prompt Evaluation. The prompt was evaluated through the prompt-sensitivity experiment described in Section 14. This experiment compares the deployed baseline prompt with controlled variants that remove the few-shot examples, change the guide style, or move the RAG context block. The objective is to measure whether prompt changes produce effects larger than ordinary model run-to-run variation, rather than relying on a single manual trial. This is especially important because small wording changes, exemplar ordering, or context placement can alter response length, refusal behaviour, and personalization strength.

11. Audio Interaction

A. Speech Synthesis. GuIA converts generated responses into speech through an audio service based on `edge-tts`. The system selects an appropriate voice for Catalan, Spanish, or English and allows the visitor to adjust the speech speed. This is particularly relevant for visitors who prefer listening to the explanation or who need more time to process spoken information.

The frontend can begin reading complete sentences while the rest of the response is still being generated. This reduces the waiting time and creates a more natural audioguide experience.

B. Speech Recognition. Visitors can also ask questions using their voice. The browser records the question and sends the audio to the backend, where OpenAI Whisper converts it into text. The transcription is shown in the interface before it is sent, allowing the visitor to review the question. This creates an alternative to keyboard interaction and makes the guide more convenient to use while moving through the museum.

12. Analytics and Administration

A. Metadata-Only Analytics. GuIA includes analytics for testing and iterative evaluation. They are disabled by default and can be activated during evaluation periods.

The system is designed to record interaction metadata rather than the text of questions or answers. The collected information can include:

1. session start and end;
2. onboarding completion;
3. preference changes;
4. locations visited;
5. question length and word count;
6. response timing;
7. response time, selected language, room, and artwork;
8. which accessibility preferences were enabled, recorded as interaction metadata rather than conversation content.

The text of visitor questions and generated answers is not stored in this analytics file. Visits are associated with session-level identifiers rather than direct personal identifiers. This information can help the museum understand how visitors use the guide while limiting the collection of unnecessary data.

B. Administration Interface. The application also includes a password-protected administration page at `/admin`. The admin area is separated from the visitor interface and checks the configured admin secret before exposing protected actions. It can report whether Neo4j and analytics are configured, display metadata-only analytics summaries, and download the raw analytics event file when analytics are enabled.

The administration interface also supports content-maintenance workflows. Museum staff can download CSV templates for supported graph entity types and upload edited files through `/admin/api/upload/<node_type>`. The backend normalizes the uploaded tables, applies the configured column mapping, and synchronizes the corresponding Neo4j nodes or relationships. The portal also exposes unresolved questions detected during use: staff can inspect pending items, reject them, or accept inferred updates when the system identifies a supported missing graph field. This keeps curatorial control with the museum rather than letting visitor questions automatically change the Knowledge Graph.

The interface is intentionally limited. It does not expose arbitrary Cypher execution to staff, and it does not store or display full visitor conversations as analytics. Its purpose is to support controlled content updates, review missing-information cases, and monitor aggregate usage patterns without requiring direct database administration.

13. Deployment

The GuIA prototype is deployed on Hugging Face Spaces using Docker. The deployed application serves the web interface, the conversational backend, and the audio services from the same public address. The iDEM-based simplification API is hosted separately and connected to the main application through an HTTP request.

Deployment is automated through GitHub Actions. When changes are added to the repository's main branch, the application is uploaded to the Hugging Face Space. API keys and database credentials are configured as private secrets and are not included in the public repository.

The deployed version is intended for testing and demonstration. A future public deployment inside the museum would require a more complete assessment of expected traffic, long-term maintenance, and institutional responsibility for the digital infrastructure.

A. Final Deliverable to the Museum. The deliverable to the Museu del Renaixement is not a single model or script, but an integrated prototype that can be tested in the museum context. The visitor-facing part is a mobile web application that runs in the browser and supports onboarding, language selection, accessibility preferences, location selection, conversational questions, voice input, and spoken answers.

The institutional part of the deliverable includes the curated Knowledge Graph, the backend retrieval and generation pipeline, the Easy-Read service integration, the metadata-only analytics mechanism, and the administration interface. These elements allow the museum to preserve control over the information used by the guide, observe general usage patterns, and update content without rewriting the visitor interface.

The prototype is also delivered as a documented software project through the GitHub repository and the Hugging Face deployment. This makes the system reproducible for demonstration and testing, while leaving clear work for a future production deployment: infrastructure planning, security review, accessibility validation with users, and long-term governance by the museum.

14. Evaluation and Testing

A. Implemented Observability. The codebase includes mechanisms that can support a formal evaluation. The system can record response times, the number of questions asked, the rooms and artworks visited, the selected language, and which accessibility preferences were active during a session. These measurements allow the team to study how visitors interact with GuIA without storing the content of their conversations by default.

B. Evaluation Framework. The project introduces a four-component evaluation framework, implemented as an automated harness and run against the deployed backend. The harness is additive and does not modify the application: it imports the backend module directly and calls its retrieval and generation functions, so the measurements reflect the same code path used during a real visit.

Every question is asked in Catalan, Spanish, and English, which allows a fair comparison across the three supported languages. The question sets are reproducible: they are sampled with a fixed random seed, so the same configuration produces the same questions on every run. Retrieval and answer generation are not deterministic, however, as both rely on a language model with no exposed temperature. Results are therefore statistical metrics over a set of questions rather than fixed per-question assertions. The retrieval evaluation includes a stability mode that re-runs each question to separate occasional variation from consistent errors.

C. Shared Evaluation Limitations. Four limitations apply across all four components and are stated here once rather than repeated after each result.

Optimistic faithfulness estimates. The faithfulness judge and the cultural-bias classifier both use the same model family as the guide (`command-a-03-2025`). A model is generally lenient when grading its own output, so faithfulness scores should be read as optimistic upper bounds and bias scores as lower bounds.

Small sample sizes. Each component covers between 60 and 90 items. The figures point to clear tendencies rather than precise values, and results should be interpreted as indicative rather than final.

Faithful does not mean correct. The graph contains a small number of noisy or incomplete entries — for example, some anonymous works have no associated author record. An answer can be judged faithful while still repeating an inaccurate value from the source. Curatorial review of the underlying data therefore remains necessary alongside any automated evaluation.

Reproducibility. The results are reproducible in method, but exact numerical reproduction requires the benchmark inputs to be archived together with the application code. The current repository contains the application code, raw museum data, and graph-construction materials; the exact benchmark harness files and question sets should be submitted as supplementary material for a fully reproducible release.

D. Generation Faithfulness. This component measures whether generated answers stay grounded in the information retrieved from the Knowledge Graph. For each question, the system retrieves museum context, generates an answer, and a second model call acts as a judge, returning a faithfulness score and a verdict classifying the answer as faithful, partially invented, or fabricated.

The question set combines three kinds of probe: grounded questions whose answer is present in the graph; out-of-graph questions about information the museum does not store; and near-miss questions that resemble answerable questions but ask for details the graph does not contain. The near-miss probes are adversarial and are designed to tempt the model into inventing a plausible answer. The full run covered 30 question specifications in each language, for a total of 90 items, all of which were judged successfully.

Across all languages, mean faithfulness was 0.95 and the fabrication rate was 4.4% (4 of 90 answers). The system refused to answer, or stated that it lacked the information, in 53% of cases, which is expected given that a large share of the question set asks for information the graph does not hold. The most important result concerns the 21 items where retrieval returned no rows at all: mean faithfulness was 1.0 and no answer was classified as a fabrication. When the system had nothing to ground its answer on, it did not invent content.

Results by question type were:

1. grounded questions: faithfulness 0.97, fabrication rate 2%;

2. out-of-graph questions: faithfulness 0.96, fabrication rate 4%;

3. near-miss questions: faithfulness 0.89, fabrication rate 11%.

Faithfulness scores were similar across languages (English 0.95, Spanish 0.97, Catalan 0.93). As expected, the adversarial near-miss questions produced the lowest scores. A representative failure is a question about international exhibitions an artwork had supposedly travelled to: the graph holds no exhibition history, and in two cases the model produced a confident but fictional list. Most such failures coincided with a retrieval error rather than with retrieved data, which is consistent with the zero-fabrication result for empty retrievals noted above.

E. Retrieval Recall. Faithfulness measures whether an answer is grounded in retrieved rows, but not whether the correct row was retrieved in the first place. An answer can be faithful even when retrieval fetched the wrong information, because the guide honestly repeated what it was given. This component addresses that gap directly.

A deterministic set of questions with known answers was built from the museum's artwork inventory, which lists 72 works together with their author, dating, technique, and room location. For each question the system performs only the retrieval step; the harness checks whether the known answer appears in the returned rows using an accent-insensitive comparison. No answer is generated and no judge is involved. The full run covered 60 question specifications per language (180 items total), of which 144 had a single expected value and 36 expected several values.

For single-valued questions, retrieval recall was 97% across all languages, with no rows returned in only 4% of cases. Recall by category was:

1. author of an artwork: 97%;

2. technique or material: 97%;

3. dating: 100%;

4. room location: 94%.

Recall across languages was 100% for English, 98% for Spanish, and 94% for Catalan. The room-location figure is a conservative lower bound: the generated query sometimes returns only the room identifier and omits the building identifier, so a correct room can be recorded as a partial match.

Multi-valued questions were harder. Pooled recall over all expected items was 62% (155 of 249 expected titles), and every expected item was returned in 81% of these questions. Retrieval found every work by a given artist (100%), but recall for all works in a room was only 58%. The main cause is that the backend caps the number of returned rows, so rooms with many works are truncated. This is a limitation of the current retrieval configuration rather than of the graph content, and it can be adjusted if needed.

F. Cultural Bias. The first two components check whether an answer is correct and grounded. This component checks whether its cultural framing is balanced. A guide can be faithful to a correct fact and still describe an artwork through a single cultural lens, omitting other influences the work actually carries. This matters for the Renaixement collection, which is genuinely mixed: a Manises plate inherits Hispano-Moresque ceramic technique, and a Flemish-school panel belongs to a Netherlandish tradition rather than an Italian one.

For each artwork, the guide is asked to describe its historical context and cultural influences. A second model call reads only the answer and distributes it across six cultural origins: Italian and Western-classical; Iberian and local; Northern and Flemish; Byzantine and Eastern; Islamic and Mediterranean; and a residual category. This gives a distribution P of what the answer emphasised. We compare P against a hand-made reference distribution Q recording the influences the work actually has, and measure the difference with the Kullback-Leibler divergence $D_{KL}(Q \parallel P)$, which we call the Cultural Bias Score. A low score means the answer matched the work; a high score means real influences were left out. Crucially, a work that is genuinely Italian and is described as Italian scores near zero: the metric reacts to divergence from the reference, not to European content as such.

This run was performed without retrieval, so the guide answered from its own internal knowledge. This separates the model's default cultural assumptions from the question of graph completeness. The full run covered 20 artworks per language (60 items total), all of which were scored.

Mean Cultural Bias Score across all languages was 2.47, falling to 1.87 once empty or evasive answers are excluded. The clearest result is the per-origin gap. The guide overweighted the Italian and Western-classical category by 0.24 and under-weighted the Northern and Flemish category by 0.44, while other categories stayed close to zero. Flemish-school paintings such as *La Visitació* and *Santa Maria Magdalena* were described as generic sixteenth-century Renaissance works with no mention of their Netherlandish character. Scores by artist origin confirm this pattern:

1. Italian works: 1.76;

2. Iberian works: 2.24;

3. Flemish works: 2.59;

4. German works: 2.67;

5. French works: 3.12.

Italian works scored lowest, which is expected in a collection where Italian influence is often the correct answer. Works of Northern, French, and German origin diverged most from their reference.

The cross-language comparison is complicated by refusals. Without retrieval, the guide often declined to answer, and refusals were most common in Spanish (70% of items, against 30% for Catalan and 10% for English). A refusal carries no cultural content but still counts as a large divergence, which inflates the raw Spanish score. When only substantive answers are counted, the order reverses and Spanish becomes the least biased of the three (1.56, against 1.98 for English and 1.85 for Catalan). The high refusal rate without context is itself informative: it shows the model is cautious about describing specific works from memory alone, which is precisely the situation the retrieval pipeline is meant to address.

The reference distributions were prepared by one annotator from a small set of works and carry a confidence level so that weaker references can be downweighted. This evaluation describes GuIA on this specific collection and should not be generalised to the model in other contexts.

G. Explanation Style and Prompt Sensitivity. The previous components evaluate the content of an answer. This component evaluates the prompt. GuIA personalises by prompting: the guide style, the visitor profile, the few-shot examples, and the ordering of prompt blocks are all prompt-level choices. It is therefore useful to know how much each choice actually changes the answer, and whether that change is a genuine effect or only run-to-run model variation.

The procedure holds everything except the prompt constant. For each question the retrieval step is run once and the resulting rows are reused across all prompt variants, ensuring that retrieval variation cannot be mistaken for a prompt effect. A baseline prompt, verified to be identical to the one used by the deployed system, is compared against three variants, each changing a single component: the few-shot examples are removed; the guide style is changed from explorer to scholar; and the RAG context block is moved from after the language rule to before it. Each prompt is run three times, providing a noise floor from the divergence between identical runs. The sensitivity ratio is the divergence between variant and baseline divided by that noise floor, measured by token overlap so the result is deterministic and requires no further model calls. A ratio near one means the change moved the answer no more than resampling would; a ratio well above one means the change had a real effect.

The two kinds of change are read in opposite directions. Removing the few-shot examples or moving the context block should not change the facts, so a ratio near one is desired. Changing the guide style is an intentional personalisation and should change the answer, so a ratio well above one is desired. The full run covered 12 grounded questions per language (each generated three times per variant). Results across all languages were:

1. few-shot examples removed: ratio 2.09;
2. guide style changed (explorer $\rightarrow$ scholar): ratio 1.28;
3. RAG context block moved: ratio 1.38.

The noise floor was 0.23.

Two findings follow. First, the few-shot examples are not a neutral choice: removing them moves the answer roughly twice as much as resampling noise, and the direction is consistent — answers produced with the examples are longer and more explanatory, while answers without them are terser, often a single sentence. The examples therefore shape the length and register of the output, not only the refusal behaviour they were included to demonstrate. Second, changing the guide style from explorer to scholar moved the answer only slightly above noise. The style instruction has a measurable but weak effect, which suggests the guide-style block is a candidate for strengthening if clearer differentiation between styles is desired. The placement of the RAG context block was close to neutral, which is the desired behaviour for a change that should not affect the facts.

The cross-language ordering is consistent: restricting the comparison to English and Spanish preserves the ranking, with the few-shot examples remaining the most influential component (ratio 1.84), the guide style the least (1.13), and the context block in between (1.43).

H. Accessibility and User Evaluation. The automated components above measure retrieval, faithfulness, cultural framing, and prompt sensitivity, but they do not answer whether GuIA works as a museum mediation device. Because accessibility is a core design objective rather than an add-on, the system also had to be evaluated through contact with the communities it is intended to serve.

Preliminary session with Down Tarragona. A preliminary exploratory session was conducted with 12 participants from Down Tarragona. The session was not designed as a statistically representative study but as a qualitative co-design activity to surface accessibility issues before deployment. Participants tested the prototype, asked questions, and provided informal feedback through a short questionnaire. The research team observed interaction problems, comprehension barriers, and reactions to interface elements such as pictograms, voice input, and simplified explanations.

Participants generally found the interface understandable, though some steps required effort. All agreed that the system would be useful and expressed interest in similar tools in other everyday contexts. The session also revealed a strong unmet need for accessible cultural mediation: only one participant had previously visited a museum. The remainder explained that they did not go to museums because they did not understand what was being shown or explained, and none had used an audioguide before. This reinforces the broader argument that accessibility barriers in museums concern the language and formats of cultural mediation, not only the usability of the digital interface.

Three concrete limitations were identified. First, the RAG pipeline was not robust to questions with many grammatical errors, incomplete words, or atypical phrasing — a relevant concern given that writing styles may differ among users with intellectual disabilities. This indicates the need for a query-normalisation layer before retrieval. Second, speech recognition produced inaccurate transcriptions for participants with atypical speech patterns. Confirmation steps or user-selectable transcription suggestions would be necessary to make voice interaction genuinely accessible. Third, some Easy-Read outputs remained too complex, suggesting that the simplification pipeline requires further co-design and validation with the intended audience before deployment.

Co-design with visual impairment specialists. Input from Asociación Discapacidad Visual Catalunya B1+B2+B3 emphasised alignment with WCAG guidelines, targeting AAA compliance where feasible. Specialists recommended full screen-reader compatibility for users who rely on assistive technologies, and integrated audio output for individuals with low vision who benefit from auditory support but do not use screen readers. These recommendations directly informed the interface and accessibility pipeline described in Section 6.

Planned living lab evaluation. A living lab at the Museu del Renaixement remains the necessary next step. The study should recruit at least 50 participants with diverse ages, language profiles, and accessibility needs, including Catalan, Spanish, and English speakers; older visitors; visitors with low digital familiarity; blind or low-vision visitors; deaf or hard-of-hearing visitors; and people with intellectual or cognitive accessibility needs.

The evaluation will combine quantitative interaction data — task completion, number of questions, selected language and accessibility options, rooms visited, response time, and session duration, collected as anonymised metadata — with qualitative post-visit questionnaires focused on perceived usefulness, clarity, ease of use, trust, and accessibility. Tasks will be structured around core interaction flows (selecting a language and profile, identifying a room or artwork, asking a question, listening to an answer, changing a setting, requesting a simpler explanation) and adapted with input from the relevant accessibility organisations so that the evaluation does not assume a single mode of interaction.

Known risks include a potentially unrepresentative sample, self-reported satisfaction masking accessibility problems, and confounding effects from smartphone unfamiliarity. These will be mitigated by combining logs, observation, questionnaires, and open feedback. Results should be read as evidence for iterative improvement rather than as a definitive validation of the system.

15. Discussion

GuIA should be understood as a mediation device rather than only as a chatbot. Its purpose is not simply to deliver information, but to support the relationship between the visitor, the museum space, and the collection. The Knowledge Graph, personalized explanations, accessibility options, and audio interaction each address a different aspect of this mediation process.

The technical results and the social innovation findings point to the same conclusion: grounding and accessibility must be designed together. The retrieval and faithfulness evaluations show that the system usually retrieves the correct factual context and rarely invents unsupported content. However, the Down Tarragona session shows that factual grounding is not enough if users cannot formulate questions in standard grammar, if speech recognition fails on atypical speech, or if simplified answers remain too complex. This means that input accessibility and output accessibility are both part of system reliability.

The cultural-bias evaluation also reframes the Knowledge Graph as more than a storage layer. The model tended to overemphasise Italian and Western-classical framing while underrepresenting Northern and Flemish influences. This suggests that future graph curation should explicitly encode cultural influences, local historical connections, and curatorial priorities, so that retrieval can counterbalance dominant model assumptions rather than merely provide isolated facts.

The retrieval results reveal a practical engineering issue as well. Single-valued factual retrieval performs strongly, but multi-valued room queries are weaker because the backend caps returned rows. This can likely be improved by changing the query strategy for broad room-level questions, for example by increasing or paginating result limits, summarising room contents before generation, or separating "overview of this room" queries from object-specific queries.

The prompt-sensitivity results are important for the personalization claim. The few-shot examples have a clear effect on length and register, but the difference between guide styles is relatively weak. This does not invalidate personalization, because language, age profile, location, audio mode, visual-description mode, and Easy-Read mode still change the interaction. It does mean that future work should strengthen the style prompts if the museum wants the guide personalities to feel clearly distinct.

The iDEM integration is intentionally bounded. It does not claim to produce formally validated Easy-Read publications automatically. It provides an accessibility-oriented conversational mode informed by an expert-developed simplification approach. This is useful for dynamic visitor questions, but the Down Tarragona findings show that output quality must still be evaluated directly with users with intellectual disabilities before stronger accessibility claims are made.

The application is designed for iterative testing. Optional analytics and the administration interface make it possible to improve the system while preserving a clear separation between museum content and the visitor interface. The broader living lab remains necessary because automatic metrics can measure retrieval and faithfulness, but they cannot fully measure museum understanding, trust, comfort, or inclusion during a real visit.

16. Limitations

Several limitations remain. First, the full living lab evaluation inside the Museu del Renaixement has not yet been conducted. The current user feedback from Down Tarragona is valuable, but it should be interpreted as exploratory and qualitative rather than statistically representative. The number of participants was limited, and the session focused specifically on users with Down syndrome rather than the full diversity of museum visitors.

Second, the planned living lab sample size of approximately 50 participants will be useful for identifying usability trends and accessibility gaps, but it will not provide enough statistical power for robust inferential comparisons between subgroups. The results should therefore be understood as evidence for iterative improvement rather than as a definitive validation of the system.

Third, the automatic evaluations rely partly on another LLM as judge. This makes large-scale evaluation feasible, but it may produce optimistic results, especially because models can be lenient when assessing outputs similar to their own. Human review remains necessary, particularly for factual accuracy, cultural framing, and accessibility.

Fourth, the cultural bias evaluation uses manually curated references prepared by the project team. Although these references are grounded in museum information and art-historical reasoning, they still involve interpretive judgement. A future evaluation should include curatorial experts and, where relevant, representatives of the cultural communities connected to the artworks.

Finally, the Down Tarragona session revealed limitations that are especially important for accessibility. The system is not yet sufficiently robust to written questions with many grammatical errors or incomplete words, and its voice recognition does not always work well with atypical speech. In addition, some Easy-Read explanations remain too complex. These findings show that accessibility cannot be assumed from the presence of accessibility features alone; it must be empirically validated with the users for whom those features are intended.

The current prototype also remains limited by the completeness of the Knowledge Graph, the quality of generated visual descriptions, ambient noise and microphone quality during speech recognition, and the infrastructure planning required before a larger public deployment.

17. Future Paths

The immediate next step is to refine the prototype before the full living lab evaluation. Based on the Down Tarragona session, three improvements are especially urgent. First, the system should include a query-normalisation layer capable of reformulating questions with spelling mistakes, incomplete words, or non-standard grammar before retrieval. This would make the RAG pipeline more tolerant of real user input. Second, the voice interaction should be improved for users with atypical speech patterns, either through better speech-recognition models, confirmation mechanisms, or alternative input options. Third, the Easy-Read pipeline should be evaluated directly with people with intellectual disabilities, ensuring that simplified answers are genuinely understandable rather than merely shorter.

The living lab evaluation within the museum will then provide empirical evidence of GuiA's performance across more diverse visitor profiles. Results should be interpreted as exploratory, given the expected sample size, but they will be crucial for identifying usability problems, accessibility gaps, and differences between visitor groups.

In the medium term, the system's language coverage could be extended to better serve the Italian and French-speaking visitors identified in the museum's visitor data. The Knowledge Graph should also be expanded and curated as both a factual and cultural document, ensuring that cultural influences, local historical connections, and interpretive priorities remain visible across all languages and personalization modes.

Finally, before broader public deployment, GuiA should undergo a formal Cultural Impact Assessment, as recommended by the CRAIF-C framework. Future work should also compare different LLMs and speech-recognition systems, not only in terms of technical accuracy, but also in relation to cultural bias, accessibility, robustness to non-standard input, and suitability for public cultural institutions.

18. Conclusion

GuIA is a deployed prototype of an adaptive and situated museum audioguide conceived as a digital mediation device. It combines a browser interface, a Neo4j Knowledge Graph, a conversational language model, multilingual interaction, voice input, spoken output, visual descriptions, and an iDEM-based Easy-Read service. Its architecture is designed to preserve the flexibility of conversation while keeping the museum's curated information at the center of the experience. In this way, the system mediates between institutional knowledge and the visitor's immediate questions, location, interests, and accessibility needs.

The project therefore provides a cautiously positive answer to the initial research question. A RAG-based conversational AI system can improve accessibility, engagement, and cultural mediation in a physical museum context, but only if it is grounded in curated institutional data, monitored by human experts, and tested with the communities it aims to serve. The preliminary Down Tarragona session shows that the need for such tools is real and that users with intellectual disabilities perceive strong value in them. At the same time, it also shows that inclusive AI requires more than adding simplified text or voice input: the system must understand imperfect questions, support atypical speech, and produce explanations that are genuinely easy to understand.

The main open challenges are therefore institutional as well as technical. The museum must continue reviewing the Knowledge Graph, validating generated explanations, and deciding how GuIA should coexist with human-guided visits. The project also needs broader user evidence from the planned living lab, stronger validation of Easy-Read and visual-description outputs, continued cultural bias audits, and a maintenance plan for any future public deployment.

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A. Reproducibility Notes

The project can be inspected and reproduced from the public repository: https://github.com/L3ms123/GuIA. The deployed demonstration is available at https://huggingface.co/spaces/RafelSV/guia-demo.

The main implementation files are:

- `app.py`: deployed Flask application and admin routes;
- `LLM/LLM_Call.py`: conversational backend, retrieval, prompting, and chat routes;
- `frontend/audio_api.py`: speech synthesis and transcription routes;
- `frontend/`: visitor interface files and JavaScript modules;

- `LLM/idem_hf_app.py`: independent iDEM-based Easy-Read service.

For local execution, the repository provides `requirements.txt`, `requirements-space.txt`, `run_guia.py`, and `run_guia.bat`. A local run can be started with `python run_guia.py` or, on Windows, `run_guia.bat`. This starts the frontend on `127.0.0.1:8000`, the audio API on `127.0.0.1:5000`, and the LLM API on `127.0.0.1:5002`.

The required configuration variables are:

- `COHERE_LLM_KEY` and `IDEM_API_URL`;
- `NEO4J_URI`, `NEO4J_USERNAME`, `NEO4J_PASSWORD`, and optionally `NEO4J_DATABASE`;
- `HF_TOKEN` for deployment;
- `WHISPER_MODEL` to control the speech-recognition model.

The museum data used to construct the Knowledge Graph are stored under `raw_data/`. The current graph construction work is documented in `KG/kg.ipynb`; query support data are stored in `KG/neo4j_query_table_data.csv`; and visual descriptions can be imported through `scripts/import_visual_descriptions.py`. Full reproduction of the graph requires the final Neo4j import procedure, graph schema, node and relationship counts, and curatorial update process described in Section 7.

The reported evaluation results require the same backend code path, the same Neo4j database content, and the same model configuration. To reproduce the exact numbers, the evaluation artifacts should be archived with the repository: the 72-work artwork inventory used for retrieval recall, the multilingual faithfulness probes, the cultural-bias reference distributions, and the prompt-sensitivity benchmark questions and variants. The evaluation should use fixed random seeds for question sampling, record the model versions used for generation and judging, and report repeated runs where retrieval or generation may vary. Without these artifacts, the report explains the evaluation methodology, but exact metric reproduction is not guaranteed.

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